

Research Interests

Database Management, Data Quality, Data Cleaning, Rule Discovery, Data Mining, Machine Learning Interpretability, Integration of Logical Reasoning and Machine Learning, Large Language Models

Education

2018.09 – 2025.06	Beihang University , Ph.D. in Computer Software and Theory <i>School of Computer Science and Engineering</i> GPA: 3.76/4, Ranked 11/91 (12%), Advisor: Prof. Wenfei Fan	Beijing, China
2014.08 – 2018.06	Xidian University , B.E. in Computer Science and Technology <i>School of Computer Science and Technology</i> GPA: 3.74/4, Ranked 3/175 (1.7%)	Xi'an, China

Publications

Note: In papers 5–8, all authors are co-authors, sorted by alphabetical order.

1. **Ziyan Han**, Jing Zhu, Jinbin Huang, Sifan Huang, Yaoshu Wang, Rui Mao*, and Jianbin Qin*. *BClean+: A Bayesian Data Cleaning System with Automated Prior Generation*. TKDE, 2026. (CCF-A; Top-tier Journal)
2. **Ziyan Han**, Wanjia Chen, Yunpeng Han, Rui Mao, and Jianbin Qin*. *Fast Diversified Top-k Rule Discovery via User-Guided Embeddings*. TKDE, 2026. (CCF-A; Top-tier Journal)
3. **Ziyan Han**, Jing Zhu, Jinbin Huang, Rui Mao*, and Jianbin Qin*. *BClean+: A Bayesian Data Cleaning System with Automated Prior Generation*. ICDE demo 2026. (CCF-A; Top-tier Conference)
4. **Ziyan Han**, Xinwei Lin, Peng Di, Chuan Xiao, Makoto Onizuka, Jiuzhang Liu, Rui Mao*, and Jianbin Qin*. *Automatic String Data Validation with Pattern Discovery*. DASFAA 2026. (CCF-B Conference)
5. Wenfei Fan, **Ziyan Han**, Min Xie, Guangyi Zhang*. *Discovering Top-k Relevant and Diversified Rules*. SIGMOD, 2025. (CCF-A; Top-tier Conference; Co-first author)
6. Wenfei Fan, **Ziyan Han**, Weilong Ren*, Ding Wang, Yaoshu Wang, Min Xie, Mengyi Yan. *Splitting Tuples of Mismatched Entities*. SIGMOD, 2024. (CCF-A; Top-tier Conference; Co-first author)
7. Wenfei Fan, **Ziyan Han**, Yaoshu Wang, Min Xie*. *Discovering Top-k Rules using Subjective and Objective Criteria*. SIGMOD, 2023. (CCF-A; Top-tier Conference; Co-first author)
8. Wenfei Fan, **Ziyan Han**, Yaoshu Wang*, Min Xie. *Parallel Rule Discovery from Large Datasets by Sampling*. SIGMOD, 2022. (CCF-A; Top-tier Conference; Co-first author)
9. Ting Deng, Lei Hou, **Ziyan Han**. *Keys as Features for Graph Entity Matching*. ICDE, 2020. (CCF-A; Top-tier)

Patents

1. Ting Deng, Lei Hou, **Ziyan Han**. *A Graph Data Entity Resolution System Based on Graph Dependency Rules and Supervised Learning*. Invention Patent, Beihang University. Granted, Patent No. ZL201911280249.1.
2. Min Xie, Guangyi Zhang, **Ziyan Han**, Wenfei Fan. *Sample Annotation Method, Rule Relevance Measurement Method, Apparatus, Device, and Medium*. Invention Patent, Shenzhen Institute of Computing Sciences. Granted, Patent No. ZL202410999366.8.
3. Min Xie, Guangyi Zhang, Wenfei Fan, **Ziyan Han**. *Method, Apparatus, Device, and Medium for Mining Entity Enhancing Rules*. Invention Patent, Shenzhen Institute of Computing Sciences. Application No. 202410938865.6. Under Substantive Examination.
4. **Ziyan Han**, Jianbin Qin, Wanjia Chen, Yunpeng Han. *A User-Guided Method, System, Terminal, and Storage Medium for Rule Embedding Learning and Diversified Evaluation*. Invention Patent, Shenzhen University. Application No.2025120481089. Under Substantive Examination.
5. **Ziyan Han**, Jianbin Qin, Wanjia Chen, Yunpeng Han. *A Diversified Top-k Rule Selection and Mining Method, System, Terminal, and Storage Medium*. Invention Patent, Shenzhen University. Application No.2025120570769.

Professional Experience

Research Assistant, Prof. Jianbin Qin's Group, Shenzhen University, Shenzhen, China 2025.07 – 2026.04

- Project 1: BClean+: A Bayesian Data Cleaning System with Automated Prior Generation (TKDE 2026; ICDE demo 2026).
- Project 2: Fast Diversified Top-k Rule Discovery via User-Guided Embeddings (TKDE 2026).
- Project 3: Automatic String Data Validation with Pattern Discovery (DASFAA 2026).

Intern Researcher, Shenzhen Institute of Computing Sciences (SICS), Shenzhen, China 2021.03 – 2024.10

- Project 1: Parallel Rule Discovery from Large Datasets by Sampling (SIGMOD 2022).
- Project 2: Discovering Top-k Rules using Subjective and Objective Criteria (SIGMOD 2023).
- Project 4: Splitting Tuples of Mismatched Entities (SIGMOD 2024).
- Project 3: Discovering Top-k Relevant and Diversified Rules (SIGMOD 2025).

Awards

2018.09 – 2025.06	Beihang University (BUAA)	
	Outstanding Graduate of Beihang University	2025.06
	SIGMOD 2025 Student Travel Grants	2025.04
	SIGMOD 2024 Student Support Scholarship	2024.04
	SIGMOD 2023 Student Travel Award	2023.04
	CASC Scholarship (Third Prize)	2022.11
	Ph.D. Second Prize Scholarship, BUAA (2020–2021)	2021.10
	Merit Student Award, BUAA (2020–2021)	2021.11
	Outstanding Graduate Student Award, BUAA (2020)	2021.01
	Merit Student Award, BUAA (2019–2020)	2020.11
	Ph.D. First Prize Scholarship, BUAA (2019–2020)	2020.09
	Merit Student Award, BUAA (2018–2019)	2019.11
	Ph.D. Second Prize Scholarship, BUAA (2018–2019)	2019.09
	Outstanding Freshman Scholarship for Ph.D. Students, BUAA	2018.09
2014.08 – 2018.06	Xidian University (XDU)	
	National Encouragement Scholarship (2015–2016)	2016.09
	Special Scholarship, XDU (2016–2017)	2017.09
	National/Provincial College Student Innovation and Entrepreneurship Training Program Certificate	2017.07
	Outstanding Student Model Award, XDU (2016–2017)	2017.11
	Outstanding Communist Youth League Member, XDU (2014–2015)	2015.05
	Second Prize Scholarship, XDU	2015.09
	Second Prize Scholarship, XDU	2014.09

Technical Skills

Python, Java, Bash, C/C++, Markdown, LaTeX, Git, Spark, etc.

Professional Service

- **External Reviewer:** AAAI 2023, APWEB 2023, TKDE 2023, ICDE 2024, ICDE 2025, APWEB 2025, WAIM 2025, DASFAA 2026.
- **Volunteer:** SIGMOD 2023.